`Effects` extends Base

$$
\begin{array}{rcl}
\tau & ::= & \ldots \\
& | & \alpha^k \mid c^k \tau \ldots \mid \tau \to^\varepsilon \tau \mid \forall \alpha^k . \tau \\
& | & \texttt{blab} \\
\\
k & ::= & * \\
& | & k \to k \\
& | & \texttt{eff} \\
& | & \texttt{lab} \\
\\
e & ::= & \ldots \\
& | & \mathsf{lambda}^\varepsilon x : \tau . e \\
& | & e[\tau] \\
& | & \texttt{handle}\ h\ e^\varepsilon \\
& | & \texttt{error}_p \\
\\
h & ::= & \{\mathsf{op} \to f, \ldots, \} \\
\\
f & ::= & \Lambda \alpha . \mathsf{lambda}^\varepsilon x : \tau_1 . \mathsf{lambda}^\varepsilon k : (\tau_2 \to^\varepsilon \tau) . e
\end{array}
$$

$$
\begin{array}{rcl}
v & ::= & \ldots \\
& | & \mathsf{lambda}^\varepsilon x : \tau . e \\
& | & \Lambda \alpha^k . v \\
& | & \texttt{handler}\ h^\varepsilon \\
& | & \texttt{perform}\ \varepsilon\ \mathsf{op}\ \tau \\
& | & \texttt{error}_p \\
\\
p & ::= & \text{Blame Party} \\
\\
E & ::= & \ldots \\
& | & E[\tau] \\
& | & \texttt{handle}^{\varepsilon(h,E)} \\
& | & \texttt{error}_E \\
\\
F & ::= & \square \\
& | & F + e \mid v + F \mid F - e \mid v - F \\
& | & F \wedge e \mid v \wedge F \mid F \vee e \mid v \vee F \\
& | & F\ e \mid v\ F \\
& | & \text{if } F \text{ then } e \text{ else } e \\
& | & F[\tau]
\end{array}
$$

Listing 1: `Effects` language